

Table 4. Spatiotemporal agreement by condition across 10,472 matched strides. GaitRite and IMU values are mean (SD), and bias = IMU minus GaitRite. Stride-length CV is calculated within participant and reported as the cohort mean.

Parameter	Condition	GaitRite	IMU	Bias	RMSE	ICC(2,1)
Gait speed (m/s)	Single-task	1.186 (0.207)	1.189 (0.215)	+0.002	0.070	0.945
	Serial 2s	1.039 (0.231)	1.043 (0.237)	+0.004	0.064	0.963
	Serial 3s	0.992 (0.229)	0.994 (0.233)	+0.002	0.062	0.964
	Serial 7s	0.994 (0.196)	0.998 (0.202)	+0.004	0.063	0.949
	Pooled	1.051 (0.230)	1.054 (0.236)	+0.003	0.065	0.961
Stride time (s)	Single-task	1.088 (0.097)	1.093 (0.110)	+0.005	0.063	0.816
	Serial 2s	1.181 (0.160)	1.177 (0.158)	-0.004	0.071	0.900
	Serial 3s	1.209 (0.168)	1.207 (0.168)	-0.002	0.071	0.911
	Serial 7s	1.218 (0.165)	1.216 (0.165)	-0.002	0.073	0.903
	Pooled	1.175 (0.159)	1.174 (0.160)	-0.000	0.070	0.905
Cadence (steps/min)	Single-task	111.2 (9.9)	110.9 (11.1)	-0.33	6.18	0.827
	Serial 2s	103.3 (12.8)	103.7 (13.2)	+0.36	5.85	0.899
	Serial 3s	101.1 (13.1)	101.3 (13.5)	+0.21	5.57	0.912
	Serial 7s	100.2 (12.5)	100.4 (12.9)	+0.18	5.67	0.900
	Pooled	103.9 (12.9)	104.0 (13.4)	+0.11	5.82	0.902
Stride-length CV (%)	Single-task	2.69	3.24	+0.56	1.12	0.504
	Serial 2s	3.18	3.39	+0.21	0.82	0.827
	Serial 3s	3.34	3.89	+0.55	1.74	0.518
	Serial 7s	3.12	3.79	+0.66	1.60	0.489
	Pooled	3.08	3.58	+0.50	1.38	0.576